

6. Limitations

Our experiments were restricted to only studying Llama-3.1-70B-Instruct. We cannot say whether the results would hold for other models under the same experiments.

Our prompts and scenarios were generated by GPT-5 mini. Their realism is not confirmed. Likewise, the elicitation rates of spontaneous deception is dependent on the quality of the scenarios, under other scenarios spontaneous deception rates might have differed.

Our analysis of spontaneous fabrication suppression used a small number of scenarios due to the small baseline elicitation of the model. Confidence intervals on the scenario flip rates from fabricative to honest under suppression steering are therefore wide, although the difference from the random steering control is large.

Deception and honesty were rated by a judge LLM, GPT-4.1 Mini. While agreement between human rating and the judge LLM was found to be favorable, it was not perfect. In some cases the judge rated omission higher than the human did simply because every detail of the ground truth was not explicitly contained in the response, despite all details necessary for the user’s query being present. The judge also sometimes treated deception types as exclusive, rating a response as only fabricative when both fabrication and omission were present.

Only results for deception as omission and as fabrication were reported in our experiments. Originally, deception as distortion was also planned to be studied, but we had difficulty producing examples of spontaneous distortion responses. Beyond distortion, omission, and fabrication, further deception varieties are possible future areas of study.

Acknowledgments

We thank Sing Hieng Wong for access to the scenario-generation notebook this paper’s dataset generation pipeline was adapted from. We thank NDIF (Fiotto-Kaufman et al., 2025) for access to their hosted models to run our experiments.

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